

ML Assignment : Polynomial Regression Report

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1 Introduction

This report details the methodology and results for developing polynomial regression models to predict continuous target variables across two unique geological and mechanical scenarios. The objective is to achieve high prediction accuracy on unseen test data by identifying the optimal polynomial degree, regularization method, and feature subsets while rigorously preventing overfitting.

2 Methodology

To ensure robust generalization, a systematic pipeline was developed to evaluate multiple configurations. The core methodology incorporates the following techniques:

- **Polynomial Expansion & Scaling:** Features were initially expanded using polynomial features. Because high-degree polynomial features exhibit massive differences in scale, standardization was applied immediately afterward to center the data (mean of 0) and scale the variance to 1, ensuring stable model convergence.
- **Comprehensive Model Grid Search:** To mitigate multicollinearity and automatically select the most predictive feature subsets, four different regression models were evaluated: Standard Linear Regression (OLS), Ridge (L2 penalty), Lasso (L1 penalty), and ElasticNet (L1 + L2 penalty).
- **Hyperparameter Tuning:** A grid search was performed over penalty values ($\alpha \in \{0.01, 0.1, 1.0, 3.0, 10.0, 30.0, 100.0\}$) for the regularized models. For ElasticNet, the mixing parameter was tested across `l1_ratio` $\in \{0.2, 0.5, 0.8\}$.
- **Cross-Validation Evaluation:** 5-Fold Cross-Validation was utilized as the primary evaluation metric. The optimal model parameters were selected based on the lowest validation Mean Squared Error (MSE), while the R^2 score was also recorded to track variance capture.

3 Phase 1: Power Plant Steam Turbine Optimization

The goal of Phase 1 was to model the Net Power Score (y) using six operational parameters representing percentage deviations.

3.1 Results and Degree Rationale

Through the optimization pipeline, the model achieved optimal performance at a polynomial degree of 5. Unlike previous single-model tests, the comprehensive grid search revealed that **Lasso Regularization** ($\alpha = 0.01$) outperformed all other models for this specific dataset.

As the degree increases from 1 to 5, the model effectively learns the underlying physical relationships of the turbine, causing the CV MSE to drop from 8.850 (at degree 1 using ElasticNet) to a definitive minimum of 0.299 at degree 5. Concurrently, the CV R^2 score peaks at 0.9686. Beyond degree 5, the MSE begins to subtly rise again (reaching 0.348 by degree 10) while the R^2 score steadily degrades. The selection of Lasso regression is highly appropriate here, as its L1 penalty enforces sparsity, mathematically reducing the coefficients of redundant 5th-degree terms to exactly zero and performing automatic feature selection.

```
--- Optimizing model for IMT2024004_train_var1.csv ---
Degree 1 | Best Model: ElasticNet(alpha=0.1, l1_ratio=0.2) | MSE: 8.850 | R2: 0.0804
Degree 2 | Best Model: Lasso(alpha=0.01) | MSE: 3.072 | R2: 0.6788
Degree 3 | Best Model: ElasticNet(alpha=0.01, l1_ratio=0.2) | MSE: 1.054 | R2: 0.8898
Degree 4 | Best Model: Lasso(alpha=0.01) | MSE: 0.700 | R2: 0.9265
Degree 5 | Best Model: Lasso(alpha=0.01) | MSE: 0.299 | R2: 0.9686
Degree 6 | Best Model: Lasso(alpha=0.01) | MSE: 0.321 | R2: 0.9663
Degree 7 | Best Model: Lasso(alpha=0.01) | MSE: 0.331 | R2: 0.9653
Degree 8 | Best Model: Lasso(alpha=0.01) | MSE: 0.338 | R2: 0.9646
Degree 9 | Best Model: Lasso(alpha=0.01) | MSE: 0.342 | R2: 0.9642
Degree 10 | Best Model: Lasso(alpha=0.01) | MSE: 0.348 | R2: 0.9635

Optimal values:
- Best Model: Lasso(alpha=0.01)
- Best Degree: 5
- Best CV MSE: 0.299
- Best CV R2: 0.9686

Saved optimal predictions to IMT2024004_pred_var1.csv
```

Figure 1: Phase 1: Terminal output confirming optimal degree and Lasso parameters.

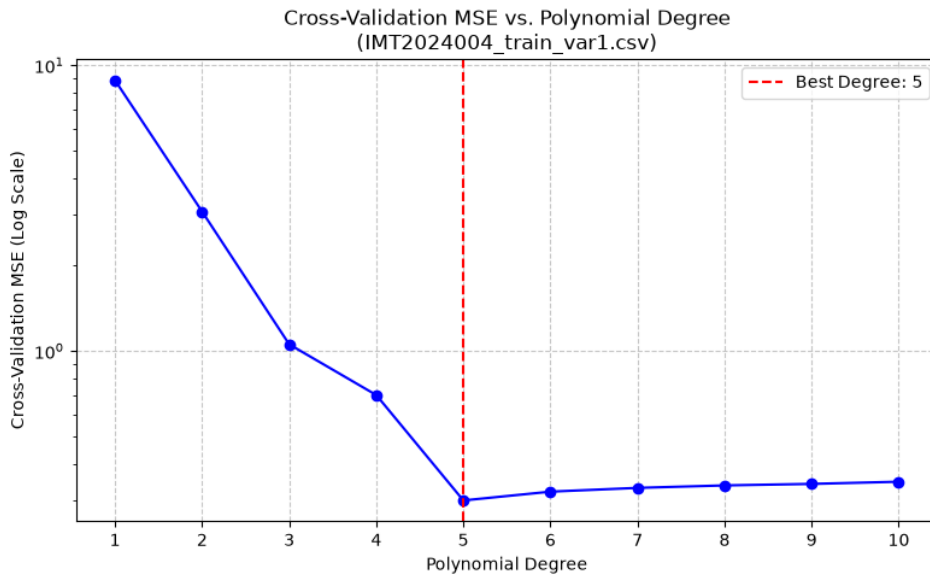


Figure 2: Phase 1: Cross-Validation MSE vs. Polynomial Degree. The red dashed line explicitly marks the optimal minimum at degree 5.

4 Phase 2: Subterranean Thermal Reservoir Mapping

The goal of Phase 2 was to predict the Thermal Anomaly Score based on spatial 3D coordinate offsets (x_1, x_2, x_3) .

4.1 Results and Degree Rationale

The geological data required a highly complex, non-linear mapping. The grid search determined that the optimal model configuration utilizes a polynomial degree of 12 combined with **Ridge Regularization** ($\alpha = 3.0$).

The CV MSE demonstrates a massive and consistent reduction from 36.901 at Degree 1 down to an absolute minimum of 0.272 at Degree 12. At this optimal configuration, the model achieves a near-perfect CV R^2 score of 0.9943. Interestingly, standard Linear Regression performs best between degrees 4 and 6, but as the degrees scale higher, L2 regularization becomes strictly necessary. After Degree 12, adding further complexity only captures noise, causing the MSE to steadily creep upward again (reaching 0.305 by degree 20 using ElasticNet). Ridge regularization ($\alpha = 3.0$) successfully controls the severe multicollinearity inherent to 12th-degree spatial expansions without aggressively dropping terms like Lasso would.

```
--- Optimizing model for IMT2024004_train_var2.csv ---
Degree 1 | Best Model: Ridge(alpha=3.0) | MSE: 36.901 | R2: 0.2356
Degree 2 | Best Model: Ridge(alpha=3.0) | MSE: 21.876 | R2: 0.5418
Degree 3 | Best Model: Ridge(alpha=1.0) | MSE: 12.148 | R2: 0.7477
Degree 4 | Best Model: LinearRegression | MSE: 4.018 | R2: 0.9153
Degree 5 | Best Model: LinearRegression | MSE: 1.532 | R2: 0.9676
Degree 6 | Best Model: LinearRegression | MSE: 0.687 | R2: 0.9855
Degree 7 | Best Model: Ridge(alpha=0.01) | MSE: 0.375 | R2: 0.9921
Degree 8 | Best Model: Ridge(alpha=0.01) | MSE: 0.300 | R2: 0.9937
Degree 9 | Best Model: Ridge(alpha=0.1) | MSE: 0.288 | R2: 0.9939
Degree 10 | Best Model: Ridge(alpha=1.0) | MSE: 0.277 | R2: 0.9942
Degree 11 | Best Model: Ridge(alpha=1.0) | MSE: 0.273 | R2: 0.9943
Degree 12 | Best Model: Ridge(alpha=3.0) | MSE: 0.272 | R2: 0.9943
Degree 13 | Best Model: Ridge(alpha=3.0) | MSE: 0.272 | R2: 0.9943
Degree 14 | Best Model: Ridge(alpha=3.0) | MSE: 0.278 | R2: 0.9941
Degree 15 | Best Model: Ridge(alpha=3.0) | MSE: 0.281 | R2: 0.9941
Degree 16 | Best Model: Ridge(alpha=3.0) | MSE: 0.294 | R2: 0.9938
Degree 17 | Best Model: Ridge(alpha=3.0) | MSE: 0.298 | R2: 0.9937
Degree 18 | Best Model: ElasticNet(alpha=0.01, l1_ratio=0.2) | MSE: 0.304 | R2: 0.9937
Degree 19 | Best Model: ElasticNet(alpha=0.01, l1_ratio=0.2) | MSE: 0.304 | R2: 0.9936
Degree 20 | Best Model: ElasticNet(alpha=0.01, l1_ratio=0.2) | MSE: 0.305 | R2: 0.9936

Optimal values:
- Best Model: Ridge(alpha=3.0)
- Best Degree: 12
- Best CV MSE: 0.272
- Best CV R2: 0.9943

Saved optimal predictions to IMT2024004_pred_var2.csv
```

Figure 3: Phase 2: Terminal output confirming the optimal degree 12 and Ridge parameters.

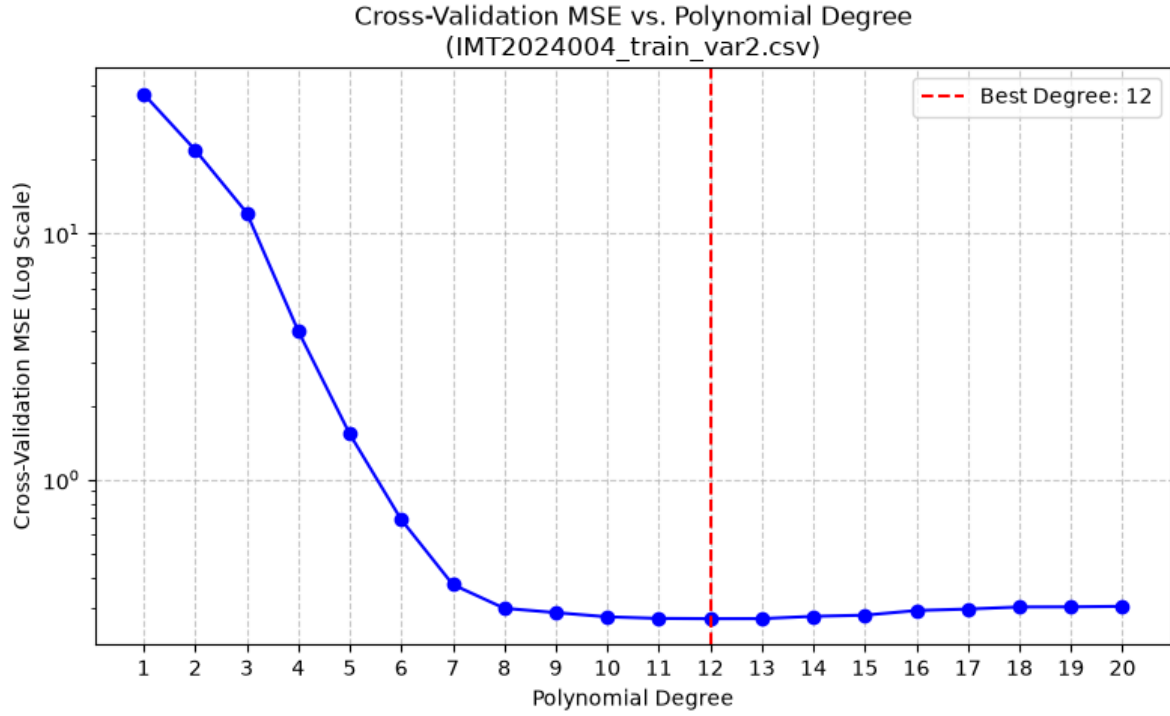


Figure 4: Phase 2: Cross-Validation MSE vs. Polynomial Degree (Log Scale). The lowest MSE validation is achieved at degree 12.

5 Conclusion

By aggressively evaluating hyperparameter combinations using 5-Fold Cross-Validation across four different regression frameworks, highly tuned models were successfully trained for both phases. The steam turbine (var1) requires a degree 5 polynomial using Lasso Regression, highlighting the dataset's need for sparse feature selection to discard irrelevant operational combinations. Conversely, the subterranean thermal map (var2) safely scales to a complex degree 12 polynomial using Ridge Regression, relying on L2 penalties to stabilize the highly collinear spatial variables. In both scenarios, tracking the CV MSE reliably identified the exact tipping point between underfitting and overfitting.

6 Deliverables

- **GitHub Repository:** https://github.com/KSaiAmith/ML_Regression.git
- **Prediction Files:** The prediction for Power Plant Steam Turbine Optimization (var1) is stored in `IMT2024004_pred_var1`, and the prediction for Subterranean Thermal Reservoir Mapping (var2) is stored in `IMT2024004_pred_var2`.
- **Model.py:** The complete regression code is stored in `model.py`.